

SecureSign – Digital Document Signature Platform Documentation

Introduction

SecureSign is a full-stack digital document signature platform inspired by modern electronic signature solutions such as DocuSign, Adobe Sign, and Zoho Sign.

The platform enables users to:

- Register and authenticate securely
- Upload PDF documents
- View uploaded documents
- Place digital signatures on documents
- Track document signing status
- Maintain complete audit logs
- Receive email notifications after signing
- Securely access protected APIs using JWT authentication

SecureSign focuses on providing a secure, scalable, and legally traceable document signing workflow while demonstrating enterprise-level backend development using Java and Spring Boot.

Design Inspiration

This project draws inspiration from leading digital signature platforms:

1. DocuSign

- Digital document workflows
- Electronic signatures
- Enterprise-grade security
- Audit trail management

2. Adobe Sign

- PDF document management
- Secure signing process

- Business document automation
- Status tracking

3. SecureSign Custom Implementation

- JWT Authentication
 - PDF Upload & Management
 - Digital Signature Placement
 - Email Notifications
 - Audit Logging
 - PostgreSQL Relational Database
 - React Dashboard
 - Spring Boot REST APIs
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Roles

User

- Register account
 - Login securely
 - Upload PDF documents
 - View uploaded documents
 - Place digital signatures
 - Track signature status
 - View audit history
 - Receive signing confirmation emails
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Use Cases of the Project

Key Use Cases

User Registration

- User creates an account
- Password is encrypted using BCrypt
- User information is securely stored

User Authentication

- User logs in
- JWT token is generated
- Protected APIs are accessed using Authorization Bearer Token

Document Management

- Upload PDF documents
- View uploaded documents
- Delete documents
- Track document status

Digital Signature Management

- Place signatures on PDFs
- Store signature coordinates
- Save signature metadata
- Update signing status

Audit System

- Track upload activity
- Track signature activity
- Record timestamps
- Record document lifecycle events

Email Notifications

- Send signing confirmation
 - Notify users about completed signatures
 - Support automated workflow communication
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Tech Stack

Backend

Spring Boot

- REST API development
- Dependency Injection
- Layered Architecture
- Business Logic Implementation

Spring Security

- Endpoint protection
- Authentication management
- Authorization handling
- Secure API access

JWT (JSON Web Token)

- Stateless authentication
- Secure API communication
- Authorization header validation
- Token-based access control

Spring Data JPA (Hibernate)

- ORM implementation
- Repository abstraction
- Entity management
- Database operations

PostgreSQL

Relational database storing:

- Users
- Documents
- Signatures
- Audit Logs

Java Mail Sender

- Email notifications
- Signature confirmation emails
- Workflow communication

Apache PDFBox

- PDF processing
- Signature placement
- PDF manipulation
- Signed document generation

Frontend

React

- Component-based architecture
- Dynamic UI rendering
- API integration
- State management

Tailwind CSS

- Utility-first styling
- Responsive layouts
- Modern dashboard interface
- Faster UI development

Axios

- REST API communication
 - JWT header injection
 - Request interceptors
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Authentication

Authentication is implemented using JWT (JSON Web Token).

Flow

User Login Request

POST /api/auth/login

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Backend validates credentials

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JWT Token Generated

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Authorization: Bearer <JWT_TOKEN>

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Protected APIs accessed securely

Security Layers

JwtFilter

- Validates JWT tokens
- Extracts authenticated users
- Secures protected endpoints

Spring Security

- Authorization management
- Endpoint security
- Authentication context

BCrypt Password Encoder

- Password hashing
 - Secure credential storage
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Flowcharts

Backend Flow

Client



JWT Authentication



Document Upload



PDF Validation



Signature Placement



Business Logic Processing



PostgreSQL Storage



Email Notification



Response Returned

Authentication Flow

Client



Register/Login Request



Credential Validation



JWT Token Generation



Token Returned



Authorization Header



Protected API Access

Document Upload Flow

Client

↓

Select PDF

↓

Multipart File Upload

↓

Spring Boot Upload API

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Database Metadata Storage

↓

Document Available in Dashboard

Digital Signature Flow

Client

↓

Select Document

↓

Enter Signature Details

↓

Choose Page & Coordinates

↓

Save Signature

↓

Update Document Status

↓

Generate Signed PDF

↓

Email Notification

Audit Flow

Client



Document Upload



Signature Action



Audit Service



Timestamp Recording



Database Storage



Audit History Response

System Architecture

Controller Layer



Handles HTTP Requests



Service Layer



Business Logic & Validation

↓

Repository Layer

↓

Database Operations

↓

PostgreSQL Storage

Database Design (ER Concept)

User

↓

uploads many Documents

↓

creates many Signatures

↓

generates many Audit Logs

Document

↓

belongs to User

↓

contains many Signatures

↓

contains many Audit Logs

Signature

↓

belongs to Document

↓

belongs to User

AuditLog

↓

belongs to Document

↓

stores signing history

PDF Processing Architecture

Client

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Multipart PDF Upload

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Spring Boot Upload API

↓

Apache PDFBox

↓

Signature Placement

↓

Generate Signed PDF

↓

Database Metadata Storage

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Frontend Preview

Frontend Integration Flow

React Frontend



Axios REST Calls



Spring Boot Backend



PostgreSQL Database



PDF Processing Service



JSON Response



Dashboard Rendering

Industry Value of the Project

This project demonstrates real-world enterprise software engineering concepts.

Industry Applications

- Digital Signature Platforms
 - Legal Document Management Systems
 - HR Contract Management
 - Banking & Financial Agreements
 - Government Approval Systems
 - Enterprise Workflow Automation
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Key Concepts Implemented

- JWT Authentication & Authorization
 - Spring Security
 - RESTful API Development
 - Layered Architecture
 - DTO-Based Design
 - PDF Upload & Processing
 - Digital Signature Management
 - Email Notification System
 - Audit Trail Logging
 - Relational Database Design
 - Full-Stack Application Development
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Real-world Inspiration Applied

DocuSign

- Digital document workflows
- Enterprise signing process
- Secure signature management

Adobe Sign

- PDF processing
- Electronic approval system
- Document lifecycle tracking

SecureSign combines both ideas into a single platform:

- Secure authentication
 - PDF document upload
 - Electronic signature placement
 - Audit logging
 - Email notification workflow
 - Enterprise-ready backend architecture
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Technologies Used

Backend Technologies

Java 21

- Enterprise programming language
- High performance
- Modern language features

Spring Boot

- REST API development
- Dependency Injection
- Layered architecture

Spring Security

- Authentication
- Authorization
- Endpoint protection

JWT

- Stateless authentication
- Secure communication

Spring Data JPA

- ORM support
- Repository pattern
- Database abstraction

PostgreSQL

Stores:

- Users
- Documents
- Signatures
- Audit Logs

Apache PDFBox

- Read PDF files
- Write signatures
- Generate signed documents

Java Mail Sender

- Email notifications
 - Signing confirmation
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Frontend Technologies

React

- Component-based UI
- Dynamic rendering
- API integration

Tailwind CSS

- Responsive layouts
- Modern design
- Utility-first styling

Axios

- HTTP communication
 - JWT interceptor
 - REST integration
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Integration Overview

React Frontend

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Axios REST API

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Spring Boot Backend

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Spring Security + JWT

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Service Layer



Spring Data JPA



PostgreSQL Database



Apache PDFBox



Java Mail Sender



Signed Document & Email Notification

Conclusion

SecureSign successfully demonstrates the design and implementation of an enterprise-grade digital document signature platform.

Key Achievements

- Secure JWT Authentication
- PDF Upload & Management
- Digital Signature Placement
- Signature Status Tracking
- Audit Logging System
- Email Notification Integration
- PostgreSQL Relational Database Design
- React Dashboard Integration
- Layered Software Architecture

The project draws inspiration from DocuSign and Adobe Sign while introducing a custom implementation focused on secure document workflows, electronic signatures, audit compliance, and enterprise backend development.

This makes SecureSign practical, scalable, industry-relevant, and an excellent portfolio project showcasing Spring Boot, React, PostgreSQL, Spring Security, JWT, PDF processing, email integration, and full-stack software engineering skills.

